

Name: Aysa Binte Masud (am12805), Evelyn Feng(af4276)

Instructor: Joachim Lebovits

Class: Probability and Statistics

Assignment: Final Python Project

Date: 08 May 2024

Development of a Statistical Analysis Calculator

Abstract

This report details the development of a statistical analysis calculator designed to compute key statistical measures including the mean, mode, variance, and standard deviation, and to visualize data from a CSV file. Aimed at enhancing understanding and interpretation of data, this tool serves as a practical application for academic and analytical purposes in the field of Probability and Statistics.

Introduction

Purpose of the Tool: The primary purpose of developing the statistical analysis calculator was to create a practical tool that integrates the core concepts of Probability and Statistics with real-world data analysis capabilities. This tool was designed with the dual objectives of enhancing learning and providing a hands-on application for statistical theories typically discussed in an academic setting.

Methodology

Development Environment: The statistical analysis calculator was developed using Python, a powerful and versatile programming language favored for its extensive libraries supporting data analysis and visualization. Key libraries utilized include Pandas for data manipulation, and Matplotlib for generating visualizations.

Data Handling: The calculator processes data input in the form of CSV files, a common format for data storage. Utilizing the Pandas library, this format facilitates easy data manipulation and

analysis. Prior to analysis, the data undergoes preprocessing to handle missing values and ensure compatibility with statistical calculations, such as converting data types and removing non-numeric entries.

Visualization Techniques: The tool incorporates several visualization techniques to aid in the analysis of the data. Using Matplotlib, it generates histograms to visualize the distribution of data, which helps in understanding the skewness and kurtosis of the dataset. These visualizations provide a graphical representation that complements the numerical analysis, making the data easier to interpret and analyze.